Extremal Combinatorics

Problem 1. Two players X and Y play a game as follows. There are n cards labelled $1, 2, \ldots, n$ respectively. Player X first removes a card. Then player Y removes two cards with consecutive labels. After that player X removes three cards with consecutive labels. Finally, player Y removes four cards with consecutive labels. What is the smallest value of n for which player Y can guarantee to complete both moves no matter how player X plays?

Problem 2. There are n cities in a country, and there are some two-way flights between some pairs of cities. We are planning for a trip to go from one city to another different city by flight. Suppose there are $k > 2014$ possible trips that include no more than one intermediate stop. Find the smallest possible value of n, and find the smallest possible value of k for that value of n. (Note that two trips are different if the routes are different. In particular, the trip going from A to B in one flight is different from the trip going from B to A in one flight.)

Problem 3. There are 99 identical balls, where 50 of them are made of copper, and the other 49 of them are made of zinc. There is a spectrometer test which can be applied to any two balls to determine whether they are made of the same metal or not. However, the result of the test can only be obtained on the next day. If all the tests should be performed today, what is the minimum number of tests required to determine the material of each ball?

Problem 4. There is a regular polygon with 2021 vertices. We label each vertex by a real number such that the labels of adjacent vertices differ by at most 1. Afterwards, we draw a diagonal between every pair of non-adjacent vertices whose labels differ by at most 1. What is the smallest possible number of diagonals that we draw?

Problem 5. There are n points on the plane with no three collinear. Each point is coloured red or blue such that every triangle with vertices of the same colour contains at least one point of the other colour in its interior. Find the largest possible value of n.

Problem 6. In each cell of an 8×8 board, there is at most one house. A house is *in the shade* if there is a house in each of the left, right and bottom adjacent cells. In particular, no house on the leftmost edge, the rightmost edge and the bottommost edge is in the shade. Find the maximum number of houses on the board if no house is in the shade.